VOICE ASSISTANCE FOR MULTIMEDIA CONTROL

SOFTWARE TEST PLAN — SPRINT 2

Version 1.1 | Sprint: 26–30 October 2026

Document Control

Field	Details
Project	Voice Assistance for Multimedia Control
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Team	Aarohi Jaiswal, Aaditya Vashisht, Abhay S
Version	1.1 — Sprint 2 update
Sprint dates	26–30 October 2026
Requirement change	NFR-3 is two hours under approved deviation D-5 / SRS v1.1; retain approval evidence.
Review / merge	Abhay S (team lead); only Abhay merges into main.

1. Introduction

1.1 Purpose

This document explains what we will test during Sprint 2, how we will test it, who is responsible, and what evidence we need to keep. We will record the results in docs/test-log.md and support them with real test output, logs, screenshots, or other relevant evidence.

1.2 System under test

The application runs in Python on a Windows laptop. It uses the laptop microphone, the offline Vosk speech recognizer, and Selenium to control YouTube in Google Chrome. To use it, the user says “hey media” and then gives one of seven commands within the listening window: play, pause, resume, next, previous, volume up, or volume down. The user must say the wake word again before giving another command.

1.3 Scope

- Verify wake-word detection, all seven supported commands, unsupported-command rejection, feedback, recovery and one-command-per-wake-word behavior.
- Measure response time and recognition reliability using recorded trial counts and acceptance criteria.
- Verify two-hour continuous availability and first-time-user usability.
- Verify wake-word gating and local audio processing using documented evidence.
- ESP32 portability is not demonstrated by the current Windows/Chrome build; record NFR-5 as a deviation/not verified unless a separate test is performed.

1.4 References and assumptions

- Software Requirements Specification (SRS) v1.1 for Voice Assistance for Multimedia Control (1_Software_Requirements_Specification_v1.1.pdf).
- Approved deviation D-5, recorded in SRS v1.1: the NFR-3 availability target is two hours.
- Repository test log: docs/test-log.md.
- Sprint 2 runs from 26 to 30 October 2026. Record actual dates, machines and versions for each test.

2. Test Items

- Wake-word detector and listening-window/session state.

- Vosk offline speech recognition and command mapping.
- Selenium media-control functions for YouTube in Chrome.
- Feedback, error handling, recovery and duplicate-command prevention.
- Performance, recognition reliability, availability, usability and local-processing security controls.
- Startup and microphone handling where the responsible teammate provides evidence.

3. Features to be Tested

Requirement	What we will check
FR-1	Wake word activates listening.
FR-2–FR-8	Play, pause, resume, next, previous, volume up and volume down.
FR-9	Unsupported input is rejected without a media action.
FR-10	Successful commands produce feedback.
NFR-1	Command response time within the SRS two-second criterion; record all trials and failures.
NFR-2	At least 90% correct recognition across 50 valid-command utterances in a quiet room.
NFR-3	Two continuous hours listening-ready without restart, under approved deviation D-5.
NFR-4	First-time-user usability; record users separately and disclose all hints.
SR-1	No command executes without a new wake word.
SR-2	Check local Vosk configuration and record network-observation evidence.
NFR-5	ESP32 portability is not verified in the current Windows/Chrome test environment.

4. Features outside this sprint's testing

- General conversation, internet search, smart-home commands or unsupported features.
- Full ESP32 implementation or cross-platform portability without a separate hardware test.
- YouTube/Chrome internals beyond the interface controlled by the application.
- Privacy claims beyond the specific backend, process, observation window and machine documented.

5. Test Approach

We will use unit tests to check individual functions, integration tests to check the full wake-word-to-action flow, and live tests with the microphone and YouTube open in Chrome. For every test, record the date, tester, machine, software/model version, steps, expected result, actual result, status, and evidence location. Keep failures in the record; if a retry succeeds, record it separately rather than replacing the failed attempt.

5.1 Security approach

- SR-1: give command-like speech without the wake word and confirm no media action occurs.
- SR-2: confirm Vosk is selected and record the network-observation method, process, duration, machine, connections and limitations. A single snapshot does not prove all traffic over time.

5.2 YouTube-specific behavior

Track speech recognition and the actual media action separately. A command may be recognized correctly even if YouTube does not carry it out. In that case, record the media action as failed and note anything that may have affected it, such as the current playback position when testing Previous.

6. Pass/Fail Criteria

- Mark a test as passed only when the observed result matches its expected result.
- NFR-1: report individual trial times, failures and summary statistics; do not report only successful trials.
- NFR-2: report correct recognitions out of 50. At least 45/50 (90%) are required; wrong recognitions and recognition failures remain in the denominator.
- NFR-3: only a documented two-hour continuous run without restart satisfies the requirement. A 15-minute run is preliminary only.
- NFR-4: document each first-time user and every hint. Describe the result as applying to the two observed sessions.
- Do not mark a security requirement as PASS unless there is supporting evidence. If a requirement also needs live testing, passing automated tests alone is not enough to mark it verified.

7. Suspension and Resumption

- Pause a live test if the microphone, application, Chrome, Selenium session, or media player is not working well enough to continue.
- Keep a record of any interruption and save the logs. Once the problem is fixed, run a quick check using the wake word, Play, and Pause before continuing.
- If the application restarts or stops being ready during the NFR-3 run, the continuous test does not count as complete. Record what happened, fix the issue, run a quick smoke test, and start a new timed run.

8. Test Deliverables

- This updated Software Test Plan.
- docs/test-log.md with results and evidence links.
- Automated test output and relevant live-run logs.
- Screenshots/configuration evidence where applicable.
- Defect/issue log and final sprint summary.

9. Test Environment

Component	Environment
Primary test machine	Windows laptop; built-in laptop microphone.
Secondary SR-2 machine	Abhay's MacBook Air/macOS for the specific network check already recorded; not the primary Windows environment.
Runtime	Python 3.x in project .venv; record exact version for each run.
Speech recognition	Vosk offline backend; models/vosk-model-small-en-in-0.4.
Browser/media	Google Chrome with YouTube open; Selenium controls the browser.
Entry point	main.py; record command-line options used.
Automated suite	python -m pytest tests/ -q (use -v for detailed output).
Setting	Quiet indoor room for baseline trials; record background noise or deviations.

10. Who does what

Person	Who does what
Aarohi Jaiswal	US-08, US-12, US-14, US-20; maintain test log and Test Plan; NFR-4 test; open PRs for Abhay to review/merge.
Aaditya Vashisht	US-10/NFR-3 two-hour run; US-11/NFR-1 and NFR-2 formal results; US-15 live recovery evidence; US-18 startup check; provide logs and start/end evidence.
Abhay S	Team lead; own integration/runtime/README/SRS; review evidence and PRs; merge into main; confirm requirement deviations.

All team members	Preserve evidence, report failures accurately, and never claim unperformed tests as passed.

11. Sprint 2 Schedule

Date	Planned work and evidence
Mon 26 Oct 2026	Sprint 2 testing and integration checks; retain actual run evidence.
Tue 27 Oct 2026	Live verification and documentation updates; record observed results.
Wed 28 Oct 2026	NFR-4 usability sessions; US-11 accuracy/speed run and logs as supplied.
Thu 29 Oct 2026	Collect Aaditya's US-10, US-11, US-15 and US-18 evidence; finish Test Plan/test-log PRs; help record the two-minute team video.
Fri 30 Oct 2026	Video, release preparation and sprint freeze; no new feature work planned.

12. Risks and Contingencies

Risk	Impact	Response
Background noise / unclear speech	Recognition errors or wrong command	Record exact phrase and outcome; each retry is a separate trial.
YouTube state affects Previous	Recognized command but action fails	Record recognition and action separately; note playback position and playlist state.
Selenium/browser disconnected	Media action fails	Record error feedback and recovery; preserve log.
NFR-3 run interrupted	Continuous run incomplete	Record interruption and restart a new timed run after smoke test.
ESP32 unavailable / laptop implementation	NFR-5 not verified	Record portability deviation; do not claim support without evidence.

13. Traceability Matrix

Requirement	Focus	Test case
FR-1	Wake/listen	TC-01
FR-2	Play	TC-02
FR-3	Pause	TC-03
FR-4	Resume	TC-04
FR-5	Next	TC-05
FR-6	Previous	TC-06
FR-7	Volume up	TC-07
FR-8	Volume down	TC-08
NFR-1	Response time	TC-09
SR-1	Wake-word gating	TC-10
FR-9	Unsupported-command rejection	TC-11
FR-10	Command feedback	TC-12
NFR-2 / US-11	50-utterance reliability	TC-13
NFR-3 / US-10	Two-hour availability (D-5)	TC-14
NFR-4	First-time-user usability	TC-15
SR-2	Local processing	TC-16
NFR-5	ESP32 portability	Deviation / not verified in current test environment

14. Test Cases

TC-01: Wake-word detection

Field	Details
Requirement traced	FR-1
Preconditions	App idle; microphone available.
Test steps	Say “hey media”; observe state.
Expected result	Enters LISTENING within 1 second.
Evidence to keep	Record state and timing.

TC-02: Play

Field	Details
Requirement traced	FR-2
Preconditions	YouTube open; media stopped/paused.
Test steps	Say wake word, then “play”.
Expected result	Playback starts within 2 seconds.
Evidence to keep	Record visible playback and time.

TC-03: Pause

Field	Details
Requirement traced	FR-3
Preconditions	Media playing.
Test steps	Say wake word, then “pause”.
Expected result	Playback pauses within 2 seconds.
Evidence to keep	Record visible pause and time.

TC-04: Resume

Field	Details
Requirement traced	FR-4
Preconditions	Media paused.
Test steps	Say wake word, then “resume”.
Expected result	Playback resumes within 2 seconds.
Evidence to keep	Record visible playback and time.

TC-05: Next

Field	Details
Requirement traced	FR-5
Preconditions	Playlist has a next item.
Test steps	Say wake word, then “next”.
Expected result	Advances to next item within 2 seconds.
Evidence to keep	Record visible item change and time.

TC-06: Previous

Field	Details
Requirement traced	FR-6
Preconditions	Playlist open; note playback position.
Test steps	Say wake word, then “previous”.
Expected result	Returns to previous item within 2 seconds, subject to player behavior.
Evidence to keep	Record recognition and action separately; note YouTube behavior.

TC-07: Volume up

Field	Details
Requirement traced	FR-7
Preconditions	Known volume level.
Test steps	Say wake word, then “volume up”.
Expected result	Volume increases by the implemented step within 1 second.
Evidence to keep	Record visible change and time.

TC-08: Volume down

Field	Details
Requirement traced	FR-8
Preconditions	Known volume level.
Test steps	Say wake word, then “volume down”.
Expected result	Volume decreases by the implemented step within 1 second.
Evidence to keep	Record visible change and time.

TC-09: Response time

Field	Details
Requirement traced	NFR-1
Preconditions	Quiet/normal low-noise setting.
Test steps	Measure end-of-utterance to action for all seven commands across planned trials.
Expected result	Meet the SRS two-second criterion; record failed actions too.
Evidence to keep	Record every time, failure and summary statistic.

TC-10: Wake-word gating

Field	Details
Requirement traced	SR-1
Preconditions	App ready/idle.
Test steps	Speak command-like words without “hey media”.
Expected result	No media action occurs without a new wake word.
Evidence to keep	Record phrases and observed state/action.

TC-11: Unsupported-command rejection

Field	Details
Requirement traced	FR-9
Preconditions	App ready.
Test steps	After wake word, try unsupported phrases; record exact phrases.
Expected result	No unsupported action executes; rejection occurs safely.
Evidence to keep	Record phrase, recognizer result, feedback and media state.

TC-12: Feedback

Field	Details
Requirement traced	FR-10
Preconditions	App and media player ready.
Test steps	Execute a supported command and observe feedback.
Expected result	Successful command produces expected visible/audible acknowledgement.
Evidence to keep	Record feedback observed.

TC-13: Recognition reliability

Field	Details
Requirement traced	NFR-2 / US-11
Preconditions	Quiet room; trial list prepared.
Test steps	Run 50 valid-command utterances across seven commands; record each result.
Expected result	At least 45/50 correctly recognized (90%).
Evidence to keep	Record all 50 outcomes; separate recognition from action success; attach log.

TC-14: Continuous availability

Field	Details
Requirement traced	NFR-3 / US-10
Preconditions	Vosk selected; mic and browser available.
Test steps	Run continuously for two hours without restart; check readiness periodically.
Expected result	Listening-ready and responsive for full two hours.
Evidence to keep	Record start/end, periodic checks, interruptions and evidence; 15 minutes is preliminary only.

TC-15: First-time-user usability

Field	Details
Requirement traced	NFR-4
Preconditions	Two first-time users; no manual.
Test steps	Ask each user to operate the app; record commands, success, hints and comments.
Expected result	Assess ability to issue working commands without manual; report assistance transparently.
Evidence to keep	Record observations separately per user.

TC-16: Local processing

Field	Details
Requirement traced	SR-2
Preconditions	Vosk offline backend selected.
Test steps	Verify backend/model; observe network traffic/connections during documented commands.
Expected result	Evidence consistent with local processing; no raw-audio transmission observed in test.
Evidence to keep	Record machine, duration, process, tools, connections and limitations.

15. Deviations and Known Limitations

D-5 — NFR-3 availability: the target is two hours, as approved by the teacher and recorded in SRS v1.1 and deviation D-5. The two-hour run passed; see the test log, Section 17.5.

NFR-5 — Portability: the current documented setup uses a Windows laptop, Python, Chrome, Selenium, and Vosk. Testing this setup does not prove that the application works on an ESP32. Keep NFR-5 marked as a deviation/not verified unless a separate ESP32 hardware test is completed.

US-11 evidence note**:** The raw log records 47 recognised command events: 40 successful actions and 7 failed Previous actions. The 40 successful actions averaged approximately 712.5 ms, with the slowest taking 1,160 ms. The raw log

does not independently establish the number of valid-command utterances or the number correctly recognised, so the claimed 45/50 accuracy result remains pending reconciliation. The Previous-command limitation is documented under deviation D-6. Recognition and media-action success are reported separately.

16. Approval and Change Control

Prepared by	Reviewer / merge authority	Status
Aarohi Jaiswal	Abhay S	Draft for review — pending PR review